

ELECTRIC VEHICLE TECHNOLOGY INTERNSHIP

Task 4: EV Design & Energy Project

Basic Electric Scooter Design and Energy Analysis

Project Type	Basic EV system design
Selected Application	Electric scooter for urban commuting
Battery System	48 V, 30 Ah lithium-ion battery
Traction Motor	BLDC motor, 2 kW rated
Target Range	Approximately 60 km per charge
Analysis	Battery, motor, range and energy consumption

Prepared as a student technical project for Task 4 of the Electric Vehicle Technology Internship.

Note: The numerical values are engineering estimates for an academic design exercise. Actual performance depends on rider mass, road conditions, tyre pressure, temperature, battery condition, motor/controller efficiency and driving style.

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1. INTRODUCTION AND DESIGN OBJECTIVE

Electric vehicles use electrical energy stored in a battery to operate an electric traction motor. The battery, battery management system, motor controller, motor, charger and auxiliary loads work together to provide vehicle propulsion. For this project, a basic electric scooter is selected because it is a practical low-voltage EV application suitable for urban travel.

The main objective is to develop a simple EV design and estimate its battery capacity, motor requirement and approximate driving range. The project also compares energy consumption for different operating conditions and provides a technical justification for the selected components.

Project objectives

- Design a basic electrical system for an urban electric scooter.
- Estimate the required battery capacity from vehicle energy demand.
- Select an appropriate traction motor rating.
- Estimate the approximate range on a full battery charge.
- Compare energy consumption under city, highway and mixed operating conditions.
- Present the final design with calculations and engineering assumptions.

Selected application

The selected vehicle is a two-wheeler electric scooter intended mainly for daily urban commuting. A moderate top speed and practical range are considered rather than high-performance operation.

2. BASIC EV SYSTEM ARCHITECTURE

The proposed system consists of a battery pack, BMS, motor controller, BLDC traction motor, charger and low-voltage auxiliary loads. The battery supplies DC electrical power to the controller. The controller regulates the current delivered to the motor according to throttle input and operating conditions.

Simplified power flow

AC Grid → Charger → 48 V Battery Pack → Motor Controller → BLDC Motor → Rear Wheel

Auxiliary path: 48 V Battery → DC-DC Converter → 12 V loads such as lighting, indicators, horn and control electronics.

Component	Main function
Battery pack	Stores electrical energy and supplies DC power.
BMS	Monitors cell voltage, temperature, current and battery protection.
Motor controller	Controls motor current, speed and torque.
BLDC motor	Converts electrical energy into mechanical torque.
Charger	Charges the battery from the AC supply.
DC-DC converter	Provides a lower DC voltage for auxiliary loads.
Throttle/control unit	Provides the rider's speed/torque command to the controller.

Working sequence

- The rider operates the throttle to request vehicle motion.
- The controller receives the throttle signal and determines the required motor current.
- The battery supplies DC power through the controller.
- The BLDC motor produces torque and drives the wheel.
- During braking, regenerative braking may return a portion of energy to the battery if supported by the system.

3. VEHICLE AND DESIGN ASSUMPTIONS

The following values are assumed for an academic first-level EV design. They are not claimed as the specifications of a particular commercial scooter.

Parameter	Assumed value
Vehicle type	Urban electric scooter
Total moving mass	180 kg (vehicle + rider + small load)
Average urban speed	35 km/h
Highway/fast-road speed	55 km/h
Rolling resistance coefficient	0.015
Aerodynamic drag coefficient	0.7
Frontal area	0.6 m ²
Air density	1.225 kg/m ³
Wheel radius	0.15 m
Drivetrain efficiency	85%
Battery nominal voltage	48 V
Battery nominal capacity	30 Ah

These assumptions allow the energy requirement to be estimated without requiring a complete vehicle simulation. The result is therefore an engineering approximation rather than a certified vehicle specification.

Design target

A nominal 48 V, 30 Ah battery is selected as the initial design. Its nominal stored energy is:

$$\text{Battery energy} = \text{Voltage} \times \text{Capacity} = 48 \times 30 = 1,440 \text{ Wh} = 1.44 \text{ kWh}$$

4. BATTERY CAPACITY ESTIMATION

Battery capacity should be sufficient to supply the expected energy demand while maintaining an operating margin. A simple energy method is used for the initial sizing.

Step 1 - Nominal battery energy

$$E_{\text{batt}} = V \times Ah = 48 \text{ V} \times 30 \text{ Ah} = 1,440 \text{ Wh}$$

Step 2 - Usable energy

For practical battery operation, 90% usable depth of discharge is assumed for this estimate.

$$E_{\text{usable}} = 1,440 \times 0.90 = 1,296 \text{ Wh}$$

Step 3 - Energy available at the wheel

Assuming an average drivetrain efficiency of 85%:

$$E_{\text{wheel}} = 1,296 \times 0.85 = 1,101.6 \text{ Wh}$$

The battery therefore provides about 1.30 kWh of usable stored energy under the chosen assumptions. After drivetrain losses, about 1.10 kWh is available as mechanical energy at the wheel side.

Battery configuration

A practical pack can be represented conceptually as a series-parallel lithium-ion arrangement selected to produce approximately 48 V nominal and 30 Ah. The exact number of cells and parallel strings depends on the chosen cell chemistry and cell rating; those details are not fixed in this academic design.

5. MOTOR REQUIREMENT AND SELECTION

A BLDC motor is selected because it provides good efficiency, compact size and suitable torque control for an electric scooter. A 2 kW rated motor is considered for the proposed urban application.

Basic road-load calculation

For a simplified estimate, the main forces are rolling resistance and aerodynamic drag.

$$\text{Rolling resistance: } F_{rr} = C_{rr} \times m \times g$$

$$F_{rr} = 0.015 \times 180 \times 9.81 \approx 26.5 \text{ N}$$

At 35 km/h (9.72 m/s), aerodynamic drag is:

$$F_d = \frac{1}{2} \rho C_d A v^2$$

$$F_d \approx 0.5 \times 1.225 \times 0.7 \times 0.6 \times 9.72^2 \approx 24.5 \text{ N}$$

$$\text{Total simplified road load} \approx 26.5 + 24.5 = 51 \text{ N.}$$

Wheel power at 35 km/h

$$P_{\text{wheel}} = F \times v \approx 51 \times 9.72 \approx 496 \text{ W.}$$

Considering drivetrain losses and additional practical loads such as acceleration, gradients and transient conditions, a motor with a much higher rating than the steady cruising requirement is selected. A 2 kW BLDC motor provides a practical reserve for acceleration and moderate slopes.

Item	Selected value	Reason
Motor type	BLDC	Efficient and suitable for electric two-wheelers
Rated power	2 kW	Provides reserve above steady cruising power
Battery	48 V, 30 Ah	Provides approximately 1.44 kWh nominal energy
Controller	48 V compatible	Matches battery and motor system

6. RANGE ESTIMATION

Range can be estimated by dividing usable battery energy by average energy consumption per kilometre.

$$\text{Range} = \text{Usable battery energy} / \text{Energy consumption per km}$$

For the selected design, a planning value of approximately 20 Wh/km is used for mixed urban operation. This includes propulsion losses and typical auxiliary consumption.

$$\text{Estimated mixed-condition range} = 1,296 \text{ Wh} / 20 \text{ Wh/km} = 64.8 \text{ km}$$

Therefore, the design target can be stated as approximately 60–65 km per charge under the assumed mixed-use conditions.

The actual range can vary significantly. Frequent acceleration, higher speed, steep gradients, low tyre pressure, heavy load and cold temperatures generally increase energy consumption.

Range sensitivity

Average consumption	Approx. range from 1,296 Wh usable energy
18 Wh/km	72 km
20 Wh/km	65 km
22 Wh/km	59 km
25 Wh/km	52 km
30 Wh/km	43 km

This sensitivity table shows why range should be expressed as an approximate value rather than a fixed distance.

7. ENERGY CONSUMPTION UNDER DIFFERENT CONDITIONS

Energy consumption changes with vehicle speed, road conditions, traffic, acceleration and load. Representative planning values are used to compare operating conditions.

Condition	Representative consumption	Estimated range
Light urban / steady riding	18 Wh/km	72 km
Normal mixed urban riding	20 Wh/km	65 km
Fast-road riding	25 Wh/km	52 km
Heavy traffic / frequent acceleration	30 Wh/km	43 km

Example energy use for a 20 km trip

At 20 Wh/km: Energy used = $20 \times 20 = 400$ Wh.

At 30 Wh/km: Energy used = $20 \times 30 = 600$ Wh.

The difference is 200 Wh for the same distance, showing the effect of operating conditions on battery usage.

Major factors affecting energy consumption

- Vehicle speed: aerodynamic drag increases rapidly as speed increases.
- Acceleration: repeated acceleration requires additional electrical power.
- Road gradient: uphill travel increases the required traction force.
- Vehicle mass: higher mass increases rolling and climbing energy requirements.
- Tyre pressure: incorrect pressure can increase rolling losses.
- Auxiliary loads: lights, displays, fans and control electronics consume additional energy.
- Regenerative braking: where available, it can recover part of the kinetic energy during deceleration.

8. FINAL DESIGN AND TECHNICAL JUSTIFICATION

The final conceptual design combines the calculated battery requirement and motor requirement into a simple electric scooter powertrain.

Subsystem	Final design value
Vehicle	Urban electric scooter
Battery	48 V, 30 Ah lithium-ion pack
Nominal battery energy	1.44 kWh
Assumed usable energy	1.296 kWh
Traction motor	2 kW BLDC
Controller	48 V BLDC motor controller
Auxiliary supply	DC-DC converter for low-voltage loads
Target mixed range	Approximately 60–65 km
Typical planning consumption	Approximately 20 Wh/km

Technical justification

- The 48 V battery provides a suitable nominal voltage for a small electric two-wheeler.
- The 30 Ah capacity gives 1.44 kWh nominal stored energy, which supports the target urban range under the stated assumptions.
- The 2 kW BLDC motor is selected with sufficient reserve above the calculated steady cruising power for acceleration and moderate road gradients.
- A BMS is required for battery protection, monitoring and safe operation.
- The controller must be matched to the battery voltage and motor current requirements.
- Actual vehicle testing is required to validate range, temperature rise, current demand and performance.

Simplified final block diagram

48 V Battery → BMS → BLDC Controller → 2 kW BLDC Motor → Wheel

48 V Battery → DC-DC Converter → Lights / Horn / Displays / Control Electronics

9. CONCLUSION

A basic electric scooter system was designed using a 48 V, 30 Ah lithium-ion battery and a 2 kW BLDC traction motor. The nominal battery energy is 1.44 kWh and the estimated usable energy, using a 90% usable fraction, is 1.296 kWh.

Using an estimated mixed-condition energy consumption of about 20 Wh/km, the calculated range is approximately 65 km. The analysis also shows that higher speed, frequent acceleration and other demanding conditions can increase energy consumption and reduce the practical range.

The project demonstrates the relationship between battery capacity, motor requirement, vehicle energy consumption and driving range. The selected values provide a reasonable starting point for a conceptual urban EV design. A real vehicle would require detailed mechanical, electrical, thermal, safety and road-testing validation before implementation.

KEY RESULTS

Parameter	Result
Battery voltage	48 V
Battery capacity	30 Ah
Nominal battery energy	1.44 kWh
Usable battery energy (assumed)	1.296 kWh
Motor rating	2 kW BLDC
Mixed-use energy assumption	20 Wh/km
Estimated range	≈ 65 km

End of Task 4 - EV Design & Energy Project